

HAT-P-37b

R-2542

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_150621 · created 2026-08-26T00:10:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457194.7226 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.102 +/- 0.04
Transit depth (Rp/Rs)^2	1.05 +/- 0.82 [%]
Orbital Inclination (inc)	87.26 +/- 2.36
Airmass coefficient 1 (a1)	1.003 +/- 0.016
Airmass coefficient 2 (a2)	-0.002 +/- 0.029
Scatter in the residuals of the lightcurve fit is	1.57 %
Best Comparison Star	#2 - [372, 429]
Optimal Aperture	2.82
Optimal Annulus	13.4
Transit Duration (day)	0.079 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.102 $\pm$ 0.04 vs 0.1394 (deviation 0.94 σ)
Duration fit vs published	0.079 d vs 0.0983 d (deviation 1.21 σ)
Mid-time O-C	-2.1 min
Depth SNR	2.5
Residual scatter	1.57 %

Light curve

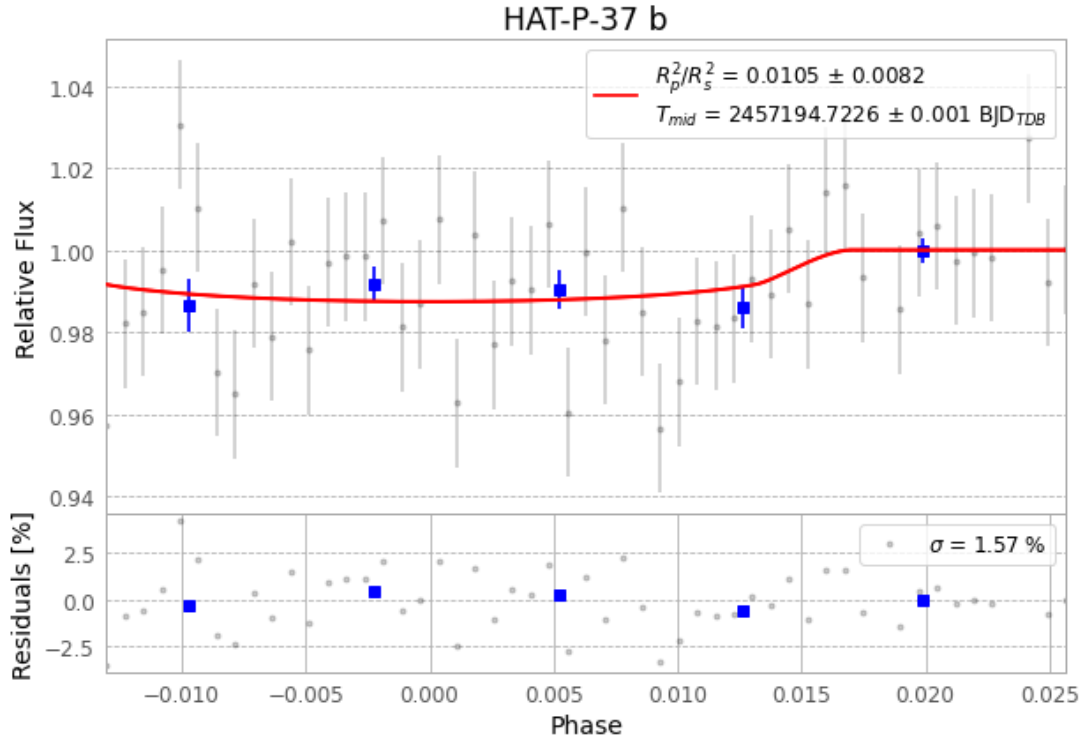


Figure 1 — FinalLightCurve_HAT-P-37 b_2015-06-20.png (SHA-256 de032007cf96ce56...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [279, 225], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8841044d0619d039...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

53 FITS frames (35 MB), HATP-37150621042412.FITS → HATP-37150621070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-150621.log (e1d59b9294c9...), FinalLightCurve_HAT-P-37 b_2015-06-20.png (de032007cf96...), FinalLightCurve_HAT-P-37 b_2015-06-20.csv (8483ce11dd93...)

Compute resources

Wall time 97.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:10Z.